Student number _____ Name _____ Major _____ Grade _____

装 订 线

Stony Brook Institute at Anhui University

The First Semester of 2020—2021 Year

《Applied Calculus II》Final Examination

(Closed book Time 120 minutes)

Question number	一	二	三	四	五	六	Total score
Score							
Marker							

一、Completions (5 items, 2 points each)

Score	

1. Let $F(x)$ be an antiderivative of $f(x)$, then $\int \sin x f(\cos x)\,dx =$ _____.

2. The tangent equation of the curve $y=\int_0^x (t-1)(t-2)\,dt$ on $(0,0)$ is _____.

3. The area of the region enclosed by $y=-x^3+x^2+2x$ and x-axis is _____.

4. Let $f(x)=\int_{\cos x}^{1} e^{-t^2}\,dt$, then $\lim\limits_{x\to 0}\dfrac{f(x)}{x^2}=$ _____.

5. The general solution of the Euler equation is $x^2\dfrac{d^2y}{dx^2}+4x\dfrac{dy}{dx}+2y=0\,(x>0)$ is _____.

二、Choices (5 items, 2 points each)

Score	

6. Suppose $f(x)>0, f'(x)<0, f''(x)>0$ on $[a,b]$,

$S_1=\int_a^b f(x)\,dx, S_2=f(b)(b-a), S_3=\dfrac{1}{2}\left[f(a)+f(b)\right](b-a)$, then ( )

A. $S_1<S_2<S_3$. B. $S_2<S_1<S_3$.

C. $S_3<S_1<S_2$. D. $S_2<S_3<S_1$.

7. Suppose the improper integrals $I_1=\int_1^{+\infty}\dfrac{dx}{x(x+1)}, I_2=\int_0^1\dfrac{dx}{x(x+1)}$, then the correct one in the following propositions is ( )

A. I_1 and I_2 are convergent. B. I_1 and I_2 are divergent.

C. I_1 is divergent and I_2 is convergent. D. I_1 is convergent and I_2 is divergent.

8. The area of the surface generated by revolving the curve $y^2 = 2px \quad (0 \le x \le a)$ about the x-axis is ( )

A. $2\pi\int_0^a \sqrt{p}\sqrt{p+2x}dx$ B. $2\pi\int_0^a \sqrt{2px}dx$

C. $2\pi\int_0^a \sqrt{2px}\sqrt{p+2x}dx$ D. $2\pi\int_0^a \frac{\sqrt{p+2x}}{\sqrt{2x}}dx$

9. Suppose $f(x)$ and $g(x)$ are continuous on $[a,b]$, and $g(x) < f(x) < m$ （m is a constant）, then the volume generated by revolving the region bounded by $y = g(x), y = f(x)$ and $x = a, x = b$ about the line $y = m$ is ( )

A. $\int_a^b \pi[2m - f(x) + g(x)][f(x) - g(x)]dx$.

B. $\int_a^b \pi[2m - f(x) - g(x)][f(x) - g(x)]dx$

C. $\int_a^b \pi[m - f(x) + g(x)][f(x) - g(x)]dx$.

D. $\int_a^b \pi[m - f(x) - g(x)][f(x) - g(x)]dx$.

10.The differential equation satisfied by the solution $y = C_1e^x + C_2e^{-2x} + xe^x$ is ( )

A. $y'' - y' - 2y = 3xe^x$. B. $y'' - y' - 2y = 3e^x$.

C. $y'' + y' - 2y = 3xe^x$. D. $y'' + y' - 2y = 3e^x$.

三、Calculations and Proofs (6 items, 8 points each)

Score	

11. Evaluate $\int \frac{1}{\sqrt{x}\left(1+\sqrt[3]{x}\right)}dx$.

12. Assume $\lim_{x\to\infty}\left(\frac{x-a}{x+a}\right)^x = \int_a^{+\infty} 4x^2 e^{-2x}dx$, find the constant a.

13. Find the length of the curve $\begin{cases} x = \int_1^t \frac{\cos u}{u}du \\ y = \int_1^t \frac{\sin u}{u}du \end{cases}, 1 \le t \le \frac{\pi}{2}$.

14. Solve the equation $\frac{dy}{dx} = \frac{-x+y-2}{x+y-4}$.

15. Suppose that $f(x)$ is a continuous function and $f(x) = \sin x - \int_0^x (x-t) f(t) dt$, show $f(x)$.

16. Let n be a positive integer, prove $\int_0^{\frac{\pi}{2}} \cos^n x \sin^n x dx = \frac{1}{2^n} \int_0^{\frac{\pi}{2}} \cos^n x dx$.

四、Analytical calculations（2 items, 8 points each）

Score	

17. Suppose the curve $y = y(x)$ through $M(1,0)$ is continuous, and the difference between the tangent slope of the curve $y = y(x)$ at $P(x, y)$ $(x \neq 0)$ and the slope of the line OP is $ax\,(a > 0)$.

(1) Find $y = y(x)$ by first order linear differential equation;

(2) The area of the region enclosed by $y = y(x)$ and $y = ax$ is $\frac{8}{3}$, find the constant a.

18. Suppose $y = y(x)$ is the solution of the differential equation $xdy + (x - 2y)dx = 0$. Find the minimum value of the volume generated by revolving the region bounded by the curve $y = y(x)$, the x-axis and the lines $x = 1, x = 2$ about the x-axis.

五、Modelling Problem（8 points）

Score	

19. For a hemispherical snowdrift, the volume melting rate is proportional to the hemisphere area S, with a constant proportion $k > 0$. Suppose that the snowdrift always remains hemisphere in the process of melting. The snowdrift with the radius r_0 has melted $7/8$ of the volume within 3 hours. How long does it take for the snowdrift to melt completely?

六、Discussion Problem（8 points）

20. We've learned a lot of concepts in 《Applied Calculus I》and《Applied Calculus II》.These concepts are the basis for some important properties and theorems. What concepts have impressed you most? Please list at least 3 concepts, choose two of them and give their definitions, and briefly explain why they impressed you.